
Language Models Reproduce Reasoning Procedures but Not Answers Under Verified Contamination

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Abstract

A benchmark item found in a model’s training data is routinely taken to mean its score on that item reflects recall, but testing the inference requires knowing what the model read, which the corpora of deployed systems do not permit. Instella publishes its full pretraining mixture and checkpoints from every stage, and its stage two corpus derives from the GSM8K training split, so membership is fixed by exact thirteen gram containment rather than estimated: against the 1319 test items, known negatives because the corpus derives only from train, the adopted threshold flags none. A premise deletion probe over 1591 train and 1588 test variants, across 891 and 888 parent problems, finds no rise in answer reproduction where the corpus enters training: a difference in differences of -0.26 points on $[-1.66, +1.14]$, holding across the later boundaries and unchanged under restriction to small answers, to terminated generations, and to a containment gradient within the train arm alone. Continued pretraining at five doses under a fixed token budget explains the null and separates two quantities usually conflated: at 64 repetitions the model reproduces 94.6 percent of injected text verbatim, yet against a never injected control the memorized solution procedure returns at $+4.76$ points $[+1.30, +8.23]$ while the final answer does not, at $+1.52$ points $[-1.93, +5.08]$. Exposure is legible in the calculation steps and not the answer. A crossed design holding sentence length, syntax and position fixed finds the main effect of an inserted unused number null at every checkpoint, with topicality null too and a residual cost of about two points for any insertion. Under a prompt that licenses declining, abstention on provably underdetermined items rises from 7.5 to 53.2 percent across the trajectory against roughly 0.1 to 2.1 percent when the option is withheld, though the strongest checkpoint still answers 44 percent of them. Two perturbation designs indicating heavy fragility did not survive verification and are reported as withdrawn. Intervals are cluster bootstraps over parent problems, 4000 draws.

1 Introduction

Finding a benchmark item in a model’s training data is, by convention, grounds for discounting the score on that item. The step from the one to the other is rarely tested, since a test needs both certainty about what the model read and access to it at more than one point in training. Neither is available for frontier systems. Detection has grown up as a literature of proxies instead, from n gram overlap [Brown et al., 2020] to perplexity based membership inference [Carlini et al., 2023, Shi et al., 2024], and plain rephrasing is enough to defeat the overlap rules [Yang et al., 2023]. All of these estimate membership. None establishes it, and an estimated treatment label cannot tell a real null apart from an uninformative one.

Instella [Liu et al., 2025] is the exception this work relies on. AMD publishes the complete pretraining mixture together with checkpoints from every stage, including the stage two synthetic mathemat-

ics corpus built from the GSM8K training split [Cobbe et al., 2021]. Most released models expose only final weights; Instella exposes the text consumed at each stage, so whether a given problem entered training, and when, becomes a matter of direct containment. Every GSM8K test item is a known negative with respect to that corpus, which turns detector error into a measured quantity: at the threshold adopted here, zero of 1319 test items are flagged, a Wilson upper bound of 0.29 percent.

Knowing an item was available to be memorized says nothing about whether memorizing it inflated the score, so the probe removes one premise the solution genuinely uses. The original answer is then no longer derivable, and a model that remembered the parent problem can still emit it while a model working from the remaining text cannot. Instella ships four checkpoints along one trajectory, stage one before the synthetic corpus appears, stage two where it enters, an instruction tuned checkpoint and a preference optimized one, so the same items go to the model on both sides of that boundary. A memorization account predicts the rate climbs once the corpus arrives. It does not. To find what the probe could have detected, selected items were injected by continued pretraining at controlled doses, and there verbatim reproduction and answer recall come apart sharply.

2 Setup

Corpus and checkpoints. The stage two corpus, `amd/Instella-GSM8K-synthetic`, ships a generated pool and a 119 thousand item subset, `train_119K`, documented as the split actually consumed in stage two pretraining. Containment is measured against that subset as a maximum over individual documents, so a claim of exposure points at one training document and not at the corpus treated as a single pooled text. Four checkpoints are probed: Stage 1, which precedes the corpus; Instella 3B, where it enters; SFT; and Instruct. Generation runs on vLLM [Kwon et al., 2023], greedy for the primary estimates, with a temperature sweep as a check.

Premise deletion. One sentence the calculator annotated solution consumes is dropped from each GSM8K train and test parent problem. Two confounds correlated with containment are handled at construction. A necessity check confirms the deleted value appears nowhere else in the prompt, and variants whose original answer survives elsewhere are thrown out, since emitting it there is copying, and that leakage runs about twice as often on high containment items. What remains is 1591 train and 1588 test rows over 891 and 888 parents. Hit rate counts variants where the model emits the deleted parent’s original answer.

Controlled injection. Calibration comes from injecting 200 verified clean GSM8K test parents, alongside 200 held out entirely, into Instella 3B by continued pretraining. The token budget is fixed across five doses of zero, one, four, sixteen and sixty four repetitions, so a higher dose displaces generic filler and does not buy extra compute. Injected document loss and verbatim continuation sit inside every arm as positive controls, since a silently failed run and a genuine absence of memorization both give a flat curve. The same probe runs on both sets at every dose.

Composition and perturbation. A depth extension probe appends one dependent calculation step to problems the model already gets right, changing the gold answer, and asks whether the step is composed or the model falls back on the original problem. Answer preserving perturbations, paraphrase, an inserted off topic sentence, entity substitution, and a sentence carrying one unused number, are each scored against the same problem’s own original.

3 Results

3.1 Recall across the training trajectory

Figure 1 gives the hit rate on GSM8K train, which sits in the stage two corpus, and on GSM8K test, absent from every corpus scanned, together with the difference in differences at each boundary. A memorization account puts the gap between stage one and stage two. Per checkpoint the train arm runs 1.13, 2.83, 3.08 and 3.58 percent against 1.64, 3.59, 3.34 and 3.15 percent on the test arm, over 1591 and 1588 rows.

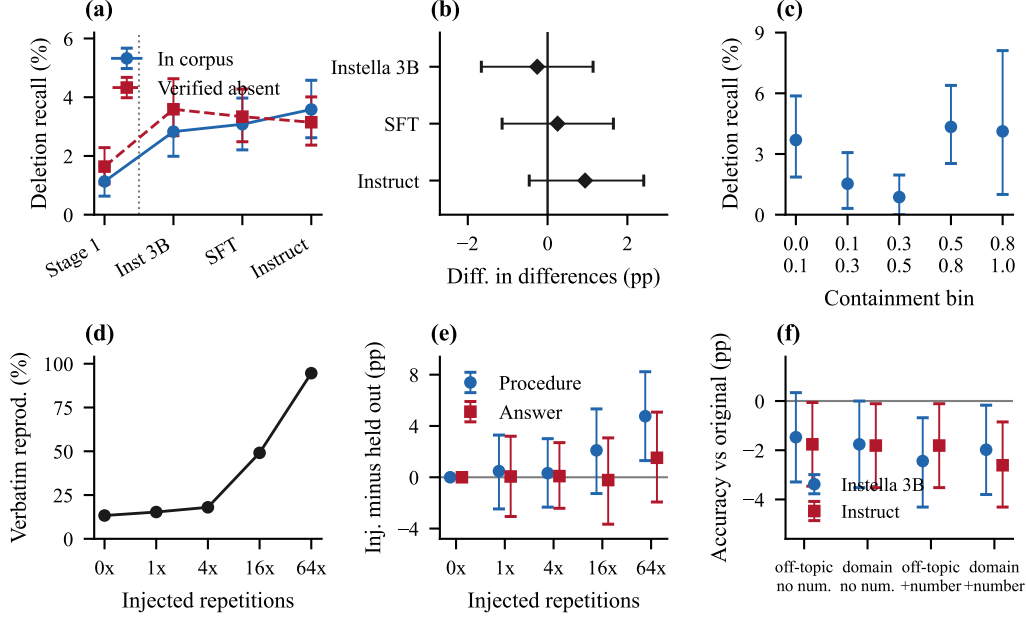


Figure 1: (a) Premise deletion recall by checkpoint for items inside the stage two corpus and for verified clean items; the dotted line marks where the corpus enters. (b) Difference in differences between the arms at each training boundary. (c) Recall within the train arm alone, binned by containment against the consumed corpus, which uses no train against test contrast. (d) Verbatim reproduction of injected documents against dose, confirming the intervention took effect. (e) Reproduction of the memorized solution procedure and of the final answer, each against the never injected control. (f) Crossed design isolating the inserted number: four twenty word insertions differing only in an off topic against a problem derived noun and a filler word against an unused integer, each against its own untouched problem. Error bars are 95 percent cluster bootstraps over parent problems, 4000 draws.

87 No gap opens at the boundary, and what movement there is runs the wrong way. The difference
88 in differences comes to -0.26 points on an interval of $[-1.66, +1.14]$, and it stays flat at every
89 later boundary too, $+0.25$ into SFT and $+0.94$ into Instruct. Three variations leave it where it is:
90 restricting to one and two digit answers, restricting to rows where both checkpoints terminated, and
91 replacing the train against test contrast with a five bin containment gradient inside the train arm
92 alone. That last one matters most, since it drops the two arm comparison entirely, and the bins run
93 3.7, 1.5, 0.9, 4.3 and 4.1 percent from least to most contained, with no order to them. Over 891
94 parent clusters the design resolves 1.89 points at 80 percent power. Every gap observed falls under
95 that.

96 3.2 Procedure and answer under controlled injection

97 Injection worked: verbatim reproduction climbs from 13.3 percent with no injection to 94.6 percent
98 at the top dose. Figure 1 sets two outcomes against the never injected control at each dose, whether
99 the deleted premise value comes back and whether the memorized intermediate calculation steps
100 come back. Procedure reproduction runs $+0.48$, $+0.32$, $+2.10$ and $+4.76$ points across the four
101 nonzero doses, the last on $[+1.30, +8.23]$; answer reproduction runs $+0.06$, $+0.09$, -0.22 and
102 $+1.52$, the last on $[-1.93, +5.08]$.

103 Answer recall never separates from the control, even where the model has all but memorized the
104 source text, while procedure reproduction climbs steadily and clears zero at the top dose. The same
105 model reproduces source text almost exactly and still fails to recover a deleted quantity or the answer
106 it determines. Memorization is legible as reproduction of calculation steps under heavy repetition,
107 not recovery of an answer, and it needs exposure far beyond what the corpus delivers to any item,
108 where repetition tops out at four.

109 3.3 Composition and surface sensitivity

110 If remembering is not the constraint, two candidates remain: failure to chain an extra step, or sen-
111 sitivity to any surface change. Neither holds. Appending one dependent step to a problem Instruct
112 already solves leaves it correct on 88.0 percent, 368 of 418, with no reversion to the pre extension
113 answer.

114 An earlier comparison suggested the inserted number was what mattered, since a sentence carrying
115 an unused quantity cost 6.2 points where a plain distractor cost 0.6. Those conditions also differed
116 in length and topicality, so a crossed design separates the three, holding length, syntax and position
117 fixed across all four cells. The main effect of the number, fixed as the primary endpoint before the
118 run, is null everywhere: -0.59 on $[-1.62, +0.45]$ at stage two, -0.79 on $[-1.93, +0.31]$ after fine
119 tuning and -0.42 on $[-1.53, +0.68]$ at Instruct, over 1764 parents each. Topicality is null too. What
120 survives is the cost of insertion itself, in panel (f) of Figure 1, where all four cells sit near -2 points
121 with overlapping intervals at stage two and at Instruct. The supervised fine tuned checkpoint is the
122 exception and shows no insertion cost at all.

123 Two points is small beside what Shi et al. [Shi et al., 2023] report for irrelevant context, and their
124 mitigation result suggests why: the frame states the record has no bearing on the question. Stripping
125 that clause costs a further -3.46 points $[-4.90, -2.01]$ at stage two but nothing after post training
126 (Appendix C).

127 3.4 Recognising an unanswerable problem

128 Every deletion item is underdetermined by construction, so declining is correct, but the prompt used
129 above demands a number and offers no alternative. Repeating the identical items under a prompt dif-
130 fering only in naming ##### unanswerable as an option changes the picture. Abstention runs 7.5,
131 11.1, 42.4 and 53.2 percent across Stage 1, Instella 3B, SFT and Instruct, against roughly 0.1, 0.0,
132 0.4 and 2.1 percent when the option is withheld, so the near zero figure measures compliance, not
133 competence, and post training buys a growing ability to recognise insufficiency. The strongest check-
134 point still answers 44 percent of problems that provably have none, every abstention is an explicit
135 refusal rather than an unparseable generation, and train and test are indistinguishable throughout.

136 3.5 Two withdrawn checks

137 Two designs looked at first like heavy evidence of fragility and neither held up, so both are reported
138 as withdrawn rather than dropped. An externally authored clause ladder appeared to cost over 30
139 points, but its later rungs are independently generated harder problems rather than the same items
140 with clauses added, so part of the drop is difficulty; Ivanova [Ivanova, 2024] makes the general form
141 of the point. Premise reordering appeared to cost over 12 points, but its generator permutes premises
142 without checking cross sentence dependency, and restricted to items free of such dependency the
143 effect lands on $+2.2$ points over $[-6.5, +10.9]$. Appendix C gives both in full.

144 4 The unavailable contrast

145 The obvious design contrasts GSM8K train items at high containment against those at low, and it was
146 tried first. Two findings closed it off, both in Appendix B. The subset the model actually consumed
147 puts 7.56 percent of train items at containment 0.80 or above against 29.75 percent computed on
148 the full released pool. More decisively, allenai/tulu-3-sft-mixture [Lambert et al., 2024]
149 sits in the stage two mixture and nvidia/OpenMathInstruct-2 [Toshniwal et al., 2024] arrives at
150 fine tuning, each carrying GSM8K train at containment 0.999 or above on over 97 percent of items
151 against under 0.1 percent for test, one of the Tulu 3 subsets involved being labelled decontaminated.
152 Every train item therefore reached the model after stage one by some route, so a high against low
153 split compares two saturated groups. The checkpoint design never sorts items by containment and
154 escapes this.

155 **Limitations and conclusion.** One family, one scale, one benchmark domain, and a dissociation
156 shown at five doses rather than ruled out at any exposure. Within those bounds verified exposure
157 does not raise how often an answer returns; it does raise how often the procedure does.

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A Related work

Huang et al. [Huang et al., 2025] construct hand authored hard perturbations of competition mathematics and attribute accuracy loss under an unchanged solution method to blind application of a learned procedure, a reading broadly consistent with the procedure level memorization in Section 3.2, established here against a corpus verified exactly rather than inferred. Wu et al. [Wu et al., 2025] show that reinforcement learning gains on mathematical reasoning depend on the pretraining exposure of the base policy and argue that clean synthetic problems are needed for reliable conclusions, a concern the checkpoint based design addresses by evaluating a policy before any relevant exposure occurred. Zhang et al. [Zhang et al., 2024] rebuild GSM8K from a contamination free procedure and report accuracy gaps of up to 13 points that track prior exposure across closed models, using an estimated exposure signal in the absence of any released training corpus; the present work substitutes an exact containment measurement for that estimate on a family where the corpus is public.

B Containment method, calibration, and corpus routes

The within split contamination contrast, high against low containment GSM8K train items, was closed rather than merely complicated by two findings. The corpus subset actually consumed during Instella 3B stage two pretraining, `train_119K`, is a 119,014 row subset of a 1,367,882 row released pool; containment computed against the pool over counts exposure, since only 7.56 percent of GSM8K train reaches 0.80 or higher against the subset actually consumed against 29.75 percent against the pool. Independently, `allenai/tulu-3-sft-mixture` is itself part of the Instella stage two pretraining mixture and `nvidia/OpenMathInstruct-2` enters at supervised fine tuning; both are generated in part from GSM8K train, and both carry it at containment 0.999 or higher on more than 97 percent of items. Every GSM8K train item therefore entered post stage one training through at least one of these routes regardless of its containment in the Instella specific synthetic corpus, so a high against low contrast defined on that corpus alone would compare two groups that are both almost entirely exposed once these mixtures are accounted for; the contrast is not attenuated by this, it is unidentifiable.

Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13 consecutive tokens, following the overlap convention of Brown et al. [Brown et al., 2020]. Containment against a document is the fraction of an item’s windows found in that document; containment against a corpus is the maximum of that fraction over every document, computed in one streaming pass with an inverted index from window to source item so cost scales linearly in corpus size. Because the corpus is generated only from the GSM8K training split, every GSM8K test item is a verified negative, and the false positive rate is measured rather than assumed: at the adopted threshold of 0.80, zero of 1319 test items are flagged, a Wilson 95 percent upper bound of 0.29 percent; a looser rule flagging any window match flags 3 of 1000 sampled test items, 0.30 percent.

Within the Tulu 3 mixture, a sampled prefix of 300,000 rows, six of the mixture’s subsets, finds GSM8K train items reaching containment 0.999 through the `flan_v2_converted` subset, 6021 of 7473 items from 89,982 rows of that source, and through `wildchat` conversation logs, 4 items from 100,000 rows, an organic route in which users pasted benchmark problems into chat. Every scan reports rows examined alongside items found, since a sampled prefix bounds containment from below and a zero means absent from the rows scanned rather than absent from the corpus.

C The irrelevance marker

The inserted frame in the crossed design ends "and has no bearing on this question", which is an instruction to ignore the insertion carried inside the insertion itself. Shi et al. [Shi et al., 2023] report that such an instruction mitigates the loss irrelevant context causes, so the modest cost measured in Section 3.3 may sit inside their mitigated regime rather than in tension with it. Testing that requires removing the clause and nothing else. The replacement, "on the afternoon of the same day", holds the sentence at exactly twenty words and preserves position, topic noun and unused integer.

The marker governs the cost at the base stage two checkpoint and not after post training. One checkpoint of three is a single result rather than a pattern, and it is reported as such: it accounts for

Table 1: Accuracy against each problem’s own original, and the effect of removing the irrelevance marker. Cluster bootstrap over parent problems, 4000 draws, 1764 parents.

Checkpoint	Marked	Unmarked	Marker effect
Instella 3B	−2.21 [−3.88, −0.51]	−5.67 [−7.43, −3.77]	−3.46 [−4.90, −2.01]
SFT	+0.09 [−1.53, +1.67]	+0.11 [−1.53, +1.81]	+0.03 [−1.33, +1.36]
Instruct	−2.21 [−3.71, −0.65]	−2.78 [−4.34, −1.10]	−0.57 [−1.93, +0.79]

the size of the stage two effect and does not account for the residual two points at Instruct, which neither the number, nor topicality, nor the marker explains.

D The two withdrawn checks in full

The clause ladder is an externally authored perturbation suite whose rungs share template identifiers, which invites pairing each perturbed item against its own template head. The instances at each rung are independently generated rather than derived from one another, so a later rung is a different and harder problem set rather than the same problems with clauses appended, and an accuracy difference across rungs therefore mixes added content with added difficulty. Premise reordering permutes the non question sentences of a problem and preserves the gold answer by construction, but a narrative word problem often carries pronouns, temporal connectives and back references across sentences, so a permutation can leave text that no reader could solve. Filtering to items where no premise after the first carries such a dependency retains 24 percent of the set, and on that subset the effect is +2.2 points on [−6.5, +10.9] over 46 problems. Manual inspection finds violations that survive the filter, so even that residual is a spot check rather than an estimate.

E Reproduction

All generation, containment, and analysis code, together with every scored generation file and containment record underlying this work, is released for audit at a public repository and a paired dataset mirror, both anonymized for review and to be de anonymized on acceptance. Every containment number reported uses the maximum over individual documents definition stated above; an earlier internal computation of the same quantity against the corpus pooled as one text produced a materially different figure for the same released split and is superseded by the document level definition throughout this work. Random seeds are fixed across item selection, deletion variant construction, the continued pretraining mixture, and every cluster bootstrap. An assistant was used to help implement generation, containment, and analysis code under the direction of the authors, who independently verified every reported number against the underlying generation files; the assistant did not design the probes or select which results to report.

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